



Sharif University Of Technology

Department of Mechanical Engineering

Fluid-Structure Interaction Project

Submitted by:

Keihan Javadi

Student Number:

400109508

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Finite Element Method for fluids

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Contents

1	Project Outline	2
2	COMSOL Setup	3
2.1	Geometry	3
2.2	Laminar Flow	4
2.2.1	Fluid Properties	4
2.2.2	Inlet and Outlet	4
2.3	Solid Mechanics	4
2.3.1	Linear Elastic Material	4
2.4	Moving Mesh	4
2.5	Mesh	5
2.6	Study 1	5
3	Results	6
3.1	Case 1: $E = 1e6$	7
3.1.1	$v = 0.01$ m/s	7
3.1.2	$v = 0.5$ m/s	8
3.1.3	$v = 0.8$ m/s	9
3.2	Case 2: $E=1e7$	10
3.2.1	$v = 0.01$ m/s	10
3.2.2	$v = 0.5$ m/s	11
3.2.3	$v = 0.8$ m/s	12
3.3	Case 3: $E=1e8$	13
3.3.1	$v = 0.01$ m/s	13
3.3.2	$v = 0.5$ m/s	14
3.3.3	$v = 0.8$ m/s	15
4	Mesh Details	16

1 Project Outline

In this project we want to calculate displacement of a solid flexible bar in a 2D rectangular channel using **Two-way Fluid-solid interaction method** in COMSOL Multiphysics software.

Here are the parameters and their values used in this simulation:

$$\text{Channel: } \begin{cases} \text{Height : 20 cm} \\ \text{Width : 5 cm} \end{cases}$$

$$\text{Fluid Properties: } \begin{cases} \text{Density : } 1000 \text{ kg/m}^3 \\ \text{Dynamic viscosity : 0.001 Pa.s} \\ \text{Fluid Velocity : [0.01, 0.5, 0.8] m/s} \end{cases}$$

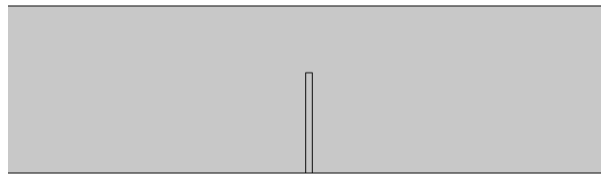
$$\text{Bar Properties: } \begin{cases} \text{Length : 3 cm} \\ \text{Thickness : 0.2 cm} \\ \text{Young Modulus : [1e6, 1e7, 1e8] Pa} \end{cases}$$

The reason for using two-way fsi is that we want to investigate the influence of fluid on the solid and vice versa.

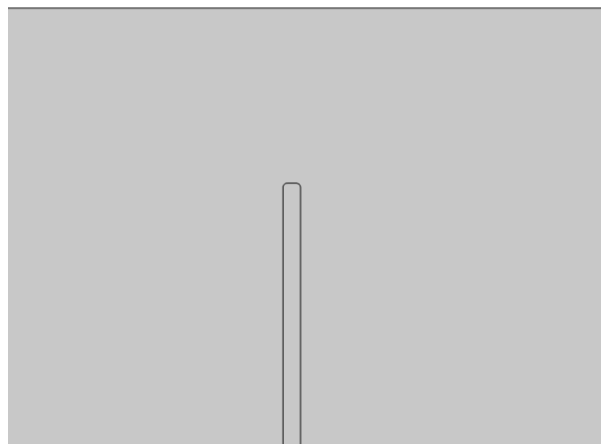
2 COMSOL Setup

2.1 Geometry

First, we start by creating geometry of channel and the flexible bar. Using **Geometry** section.



Because the tip of the bar have sharp edges, it may experience bad meshing before and after the simulation. So we apply some fillet at the top edges of the bar as below.



2.2 Laminar Flow

For **domain selection**, we choose whole domain except the bar, and we set **1 atm** for reference pressure level and **293.15 K** for reference temperature.

2.2.1 Fluid Properties

In this tab, we set Density and Dynamic viscosity as 1000 kg/m^3 and $1 \times 10^{-3} \text{ Pa.s}$.

2.2.2 Inlet and Outlet

Now we use **Inlet** and **Outlet** section in Laminar flow tab.

For inlet, we have different velocities, so we define **vel** parameter in **Global definitions\Parameters** and for initial velocity, we set it as 0.01 m/s .

For Outlet, we assume that the fluid is going into atmospheric pressure so we set 0 as pressure difference between the outlet and free space.

2.3 Solid Mechanics

In this tab, we the flexible bar as domain and in section Discretization, we set Displacement Field as **Quartic Lagrange**.

The reason we use this discretization is that if, for example we use Linear or quadratic lagrange, COMSOL can not converge the answer we face errors.

2.3.1 Linear Elastic Material

Again in this tab we choose the rod as domain and in Linear Elastic section, for Material symmetry we choose **Isotropic** and we specify it as **Young's modulus and Poisson's ratio**.

In our simulations, we set Young's modulus as **1e6**, **1e7**, **1e8** and the remains stays constant for all simulations. We set **0.35** for poisson's ratio and **1234 kg/m³** for density, respectively.

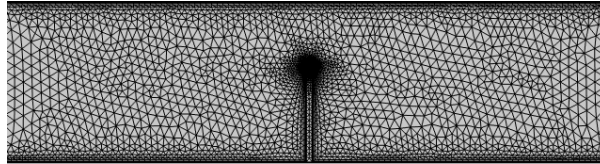
Also we want to find the deformation tip of the bar, so we need to fix end of the bar to channel.

2.4 Moving Mesh

In this simulation we want to investigate the deformation of the bar. So the initial mesh of the geometry will change during the simulation. So, in this tab we set channel as domain.

2.5 Mesh

We want to use COMSOLs default mesh systems. So we use **Physics-controlled mesh** as Sequence type, and for Element size we use **Normal**.



2.6 Study 1

Because we have different velocities, we define **Parametric Sweep** and we fill it as below.

Sweep type: All combinations		
Parameter name	Parameter value list	Parameter unit
vel	0.01 0.5 0.8	

Also in Stationary Solver, we use **Fully Coupled** method. In this method, we use **Newton** nonlinear method, and for best results, we set Maximum number of iteration to **500**.

3 Results

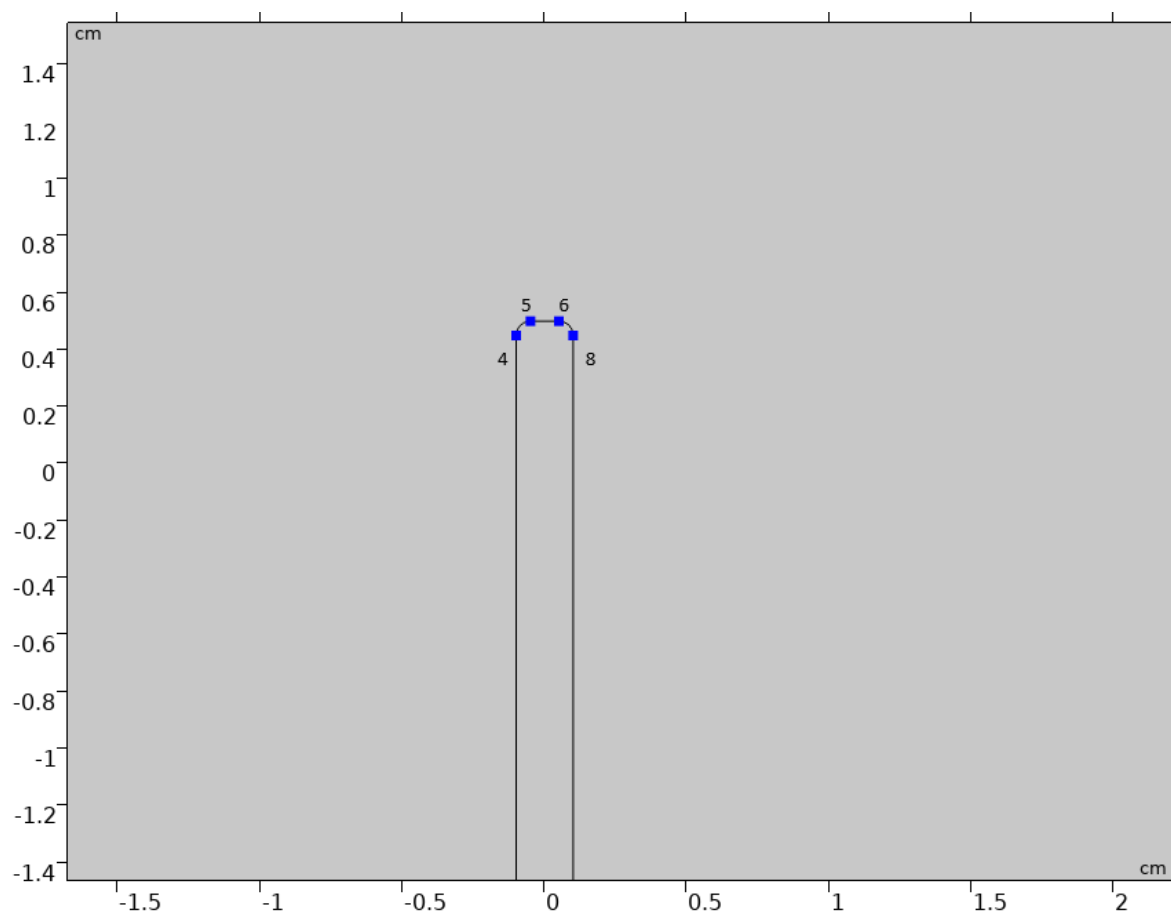


Figure 1: Points in which we checked their displacement

3.1 Case 1: $E = 1e6$

3.1.1 $v = 0.01$ m/s

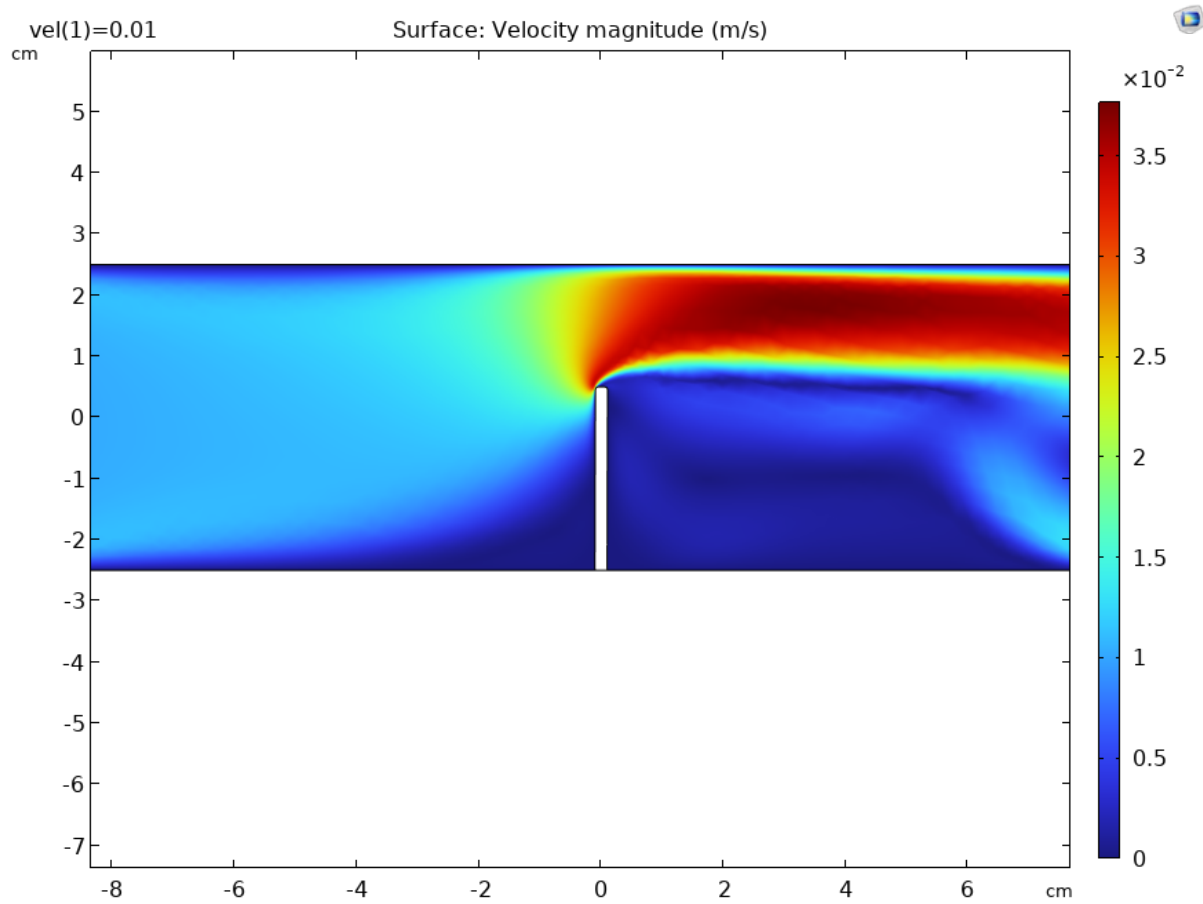


Figure 2: Velocity contour for $v=0.01$ and $E=1e6$

Velocity (m/s)	Displacement magnitude (cm)			
	Point 4	Point 5	Point 6	Point 8
0.01	0.007 410 5	0.007 569 7	0.007 569 6	0.007 410 2

Table 1: Displacement magnitudes at different points for velocity 0.01 m/s.

3.1.2 $v = 0.5$ m/s

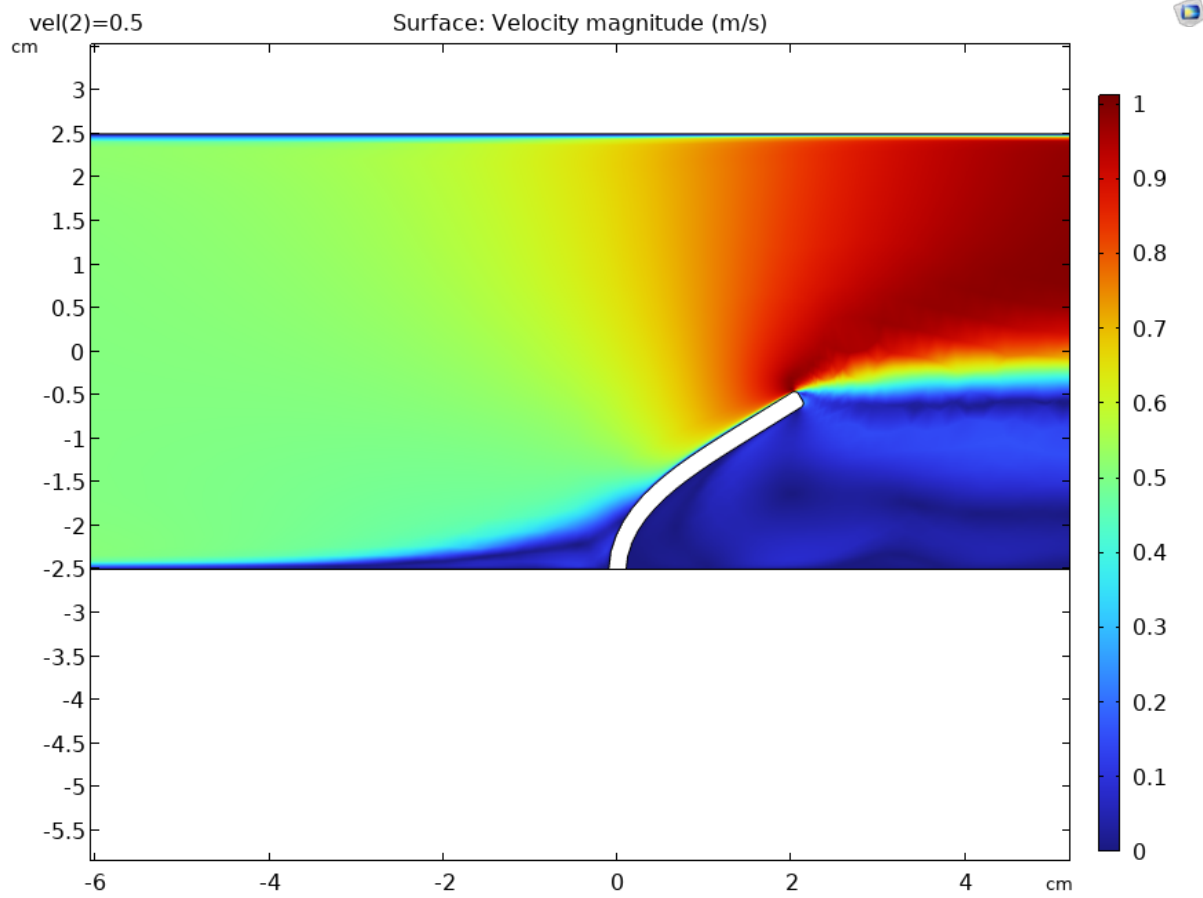


Figure 3: Velocity contour for $v=0.5$ and $E=1e6$

Velocity (m/s)	Displacement magnitude (cm)			
	Point 4	Point 5	Point 6	Point 8
0.5	2.3005	2.3447	2.3393	2.2895

Table 2: Displacement magnitudes at different points for velocity 0.5 m/s.

3.1.3 $v = 0.8$ m/s

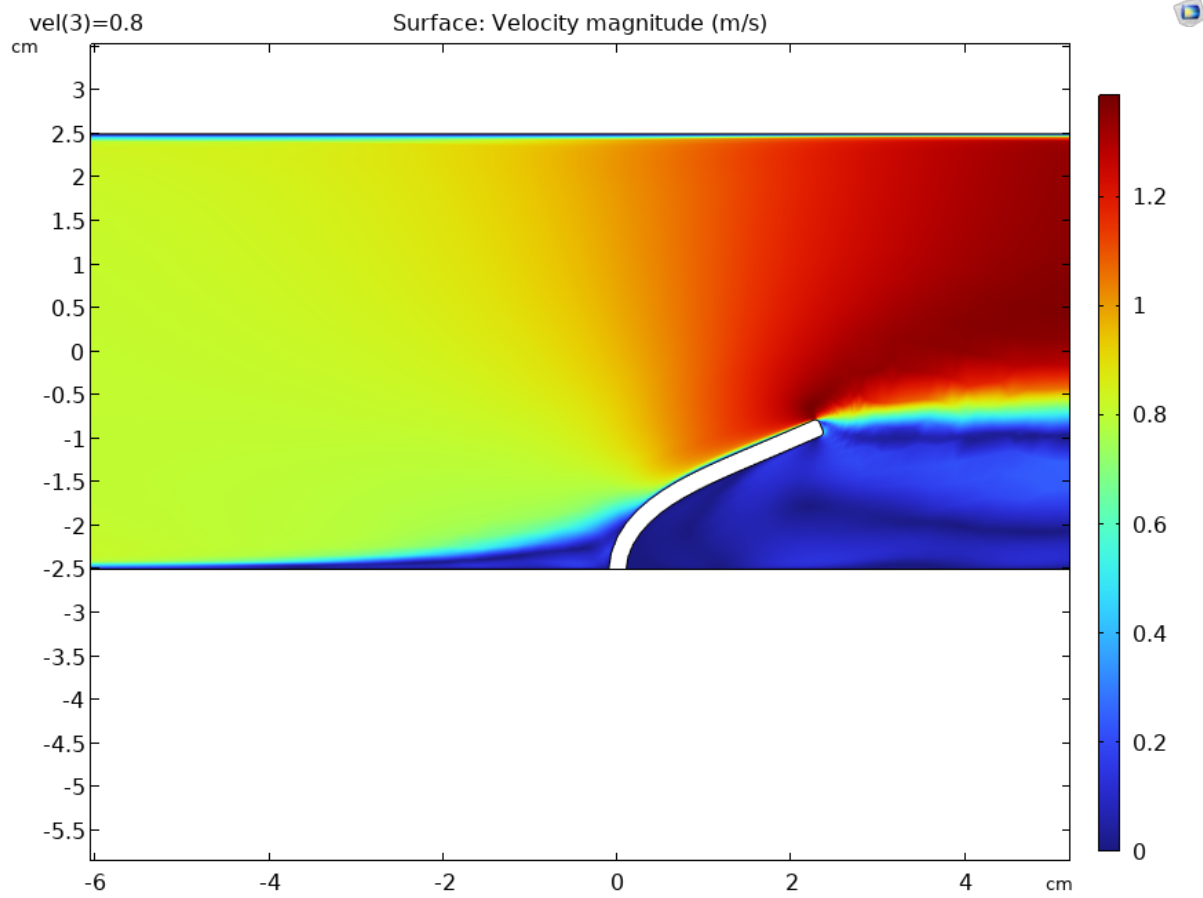


Figure 4: Velocity contour for $v=0.8$ and $E=1e6$

Velocity (m/s)	Displacement magnitude (cm)			
	Point 4	Point 5	Point 6	Point 8
0.8	2.6610	2.7110	2.7048	2.6485

Table 3: Displacement magnitudes at different points for velocity 0.8 m/s.

3.2 Case 2: $E=1e7$

3.2.1 $v = 0.01$ m/s

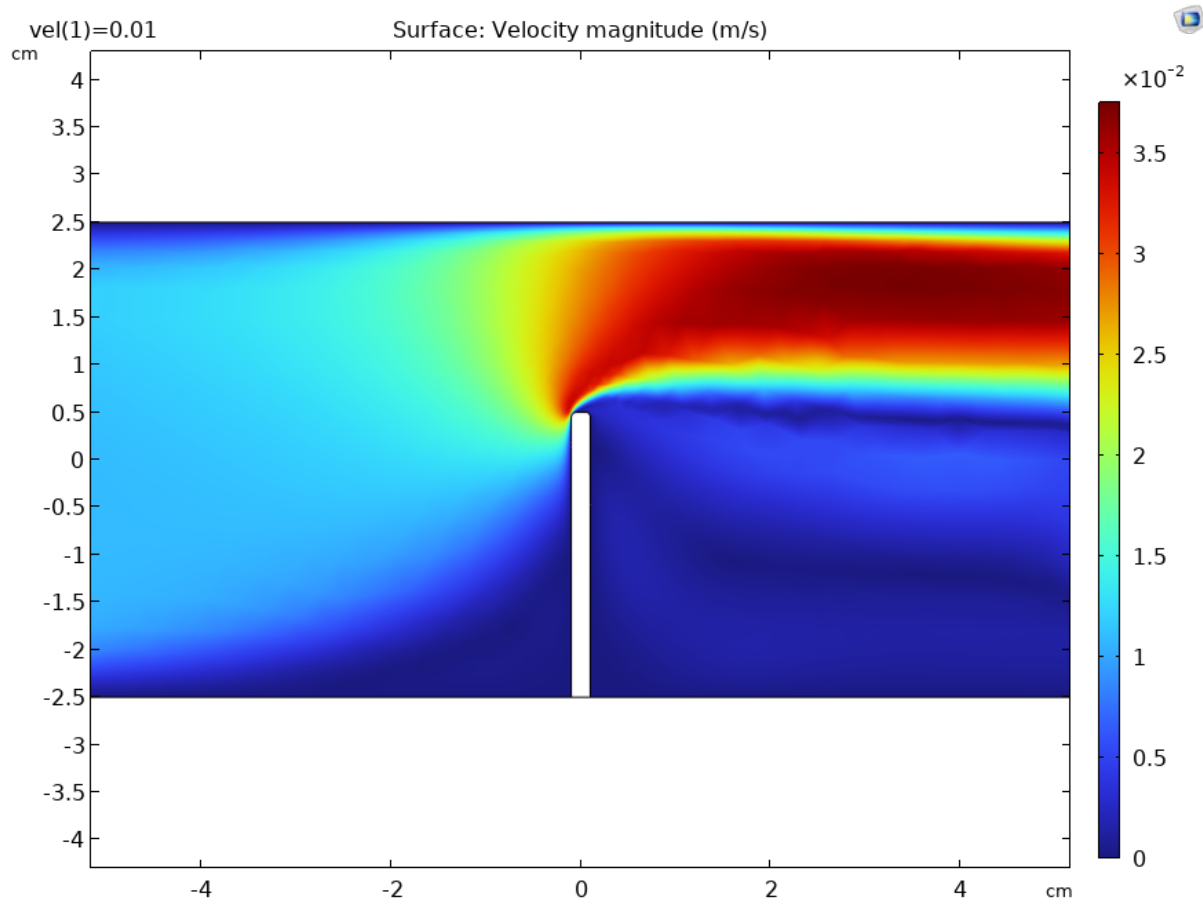


Figure 5: Velocity contour for $v=0.01$ and $E=1e7$

Velocity (m/s)	Displacement magnitude (cm)			
	Point 4	Point 5	Point 6	Point 8
0.01	0.000 742 40	0.000 758 36	0.000 758 36	0.000 742 39

Table 4: Displacement magnitudes at different points for velocity 0.01 m/s.

3.2.2 $v = 0.5 \text{ m/s}$

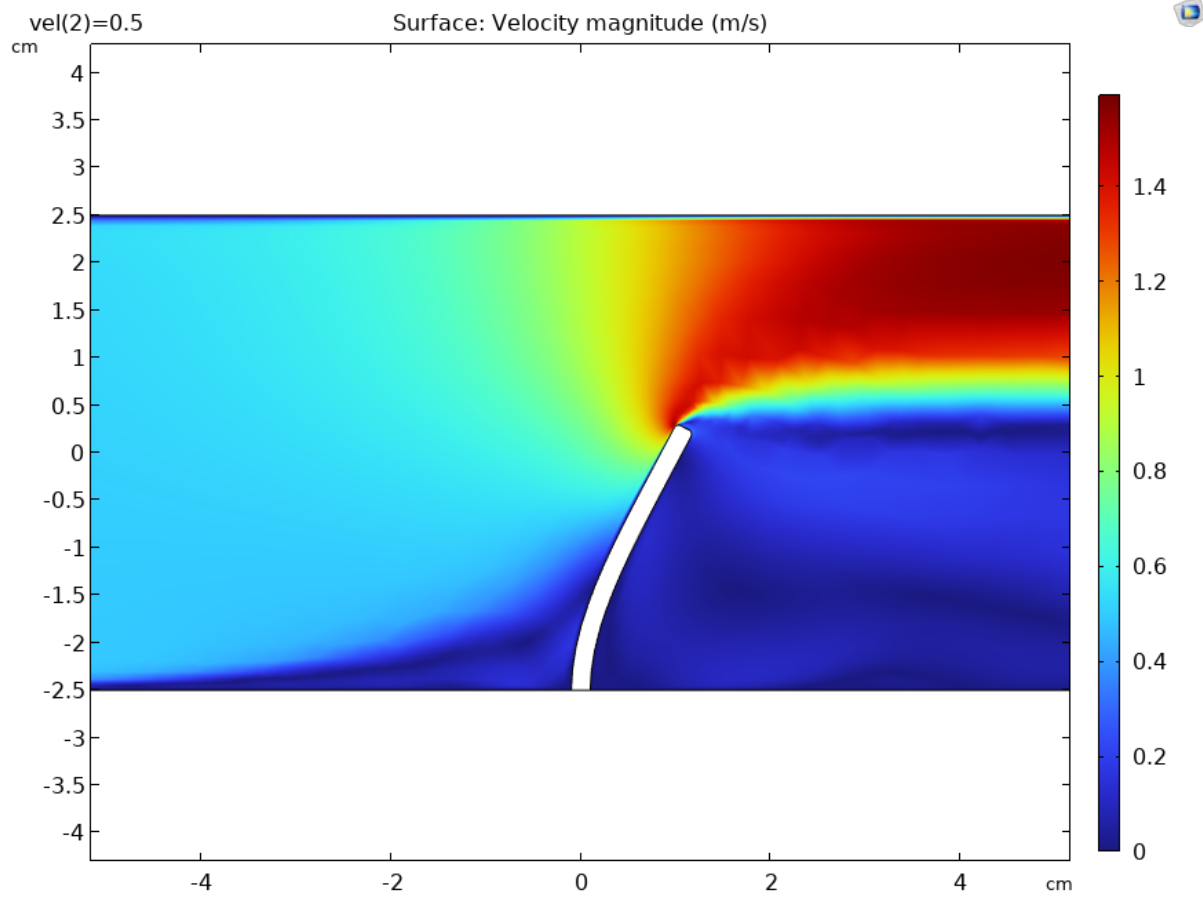


Figure 6: Velocity contour for $v=0.5$ and $E=1e7$

Velocity (m/s)	Displacement magnitude (cm)			
	Point 4	Point 5	Point 6	Point 8
0.5	1.1020	1.1246	1.1232	1.0990

Table 5: Displacement magnitudes at different points for velocity 0.5 m/s .

3.2.3 $v = 0.8 \text{ m/s}$

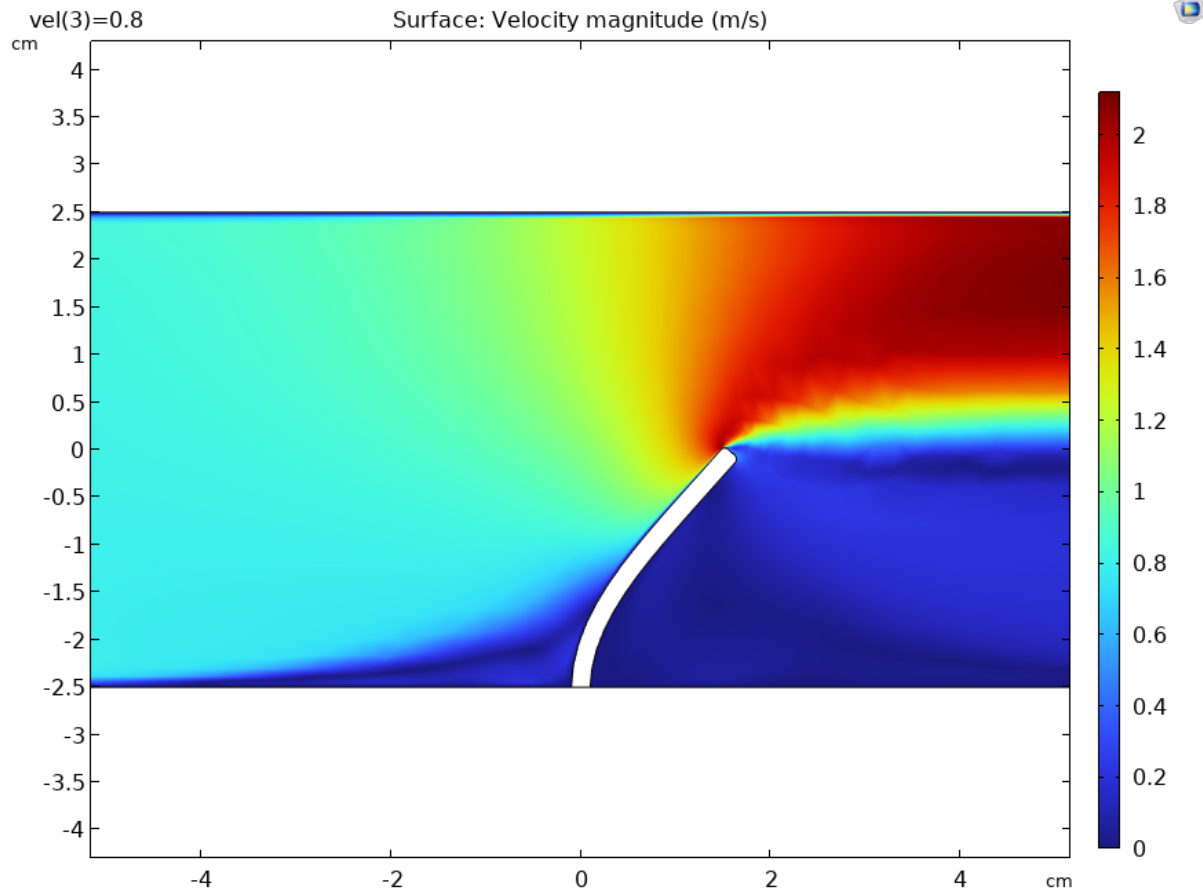


Figure 7: Velocity contour for $v=0.8$ and $E=1e7$

Velocity (m/s)	Displacement magnitude (cm)			
	Point 4	Point 5	Point 6	Point 8
0.8	1.6429	1.6758	1.6726	1.6365

Table 6: Displacement magnitudes at different points for velocity 0.8 m/s .

3.3 Case 3: $E=1e8$

3.3.1 $v = 0.01$ m/s

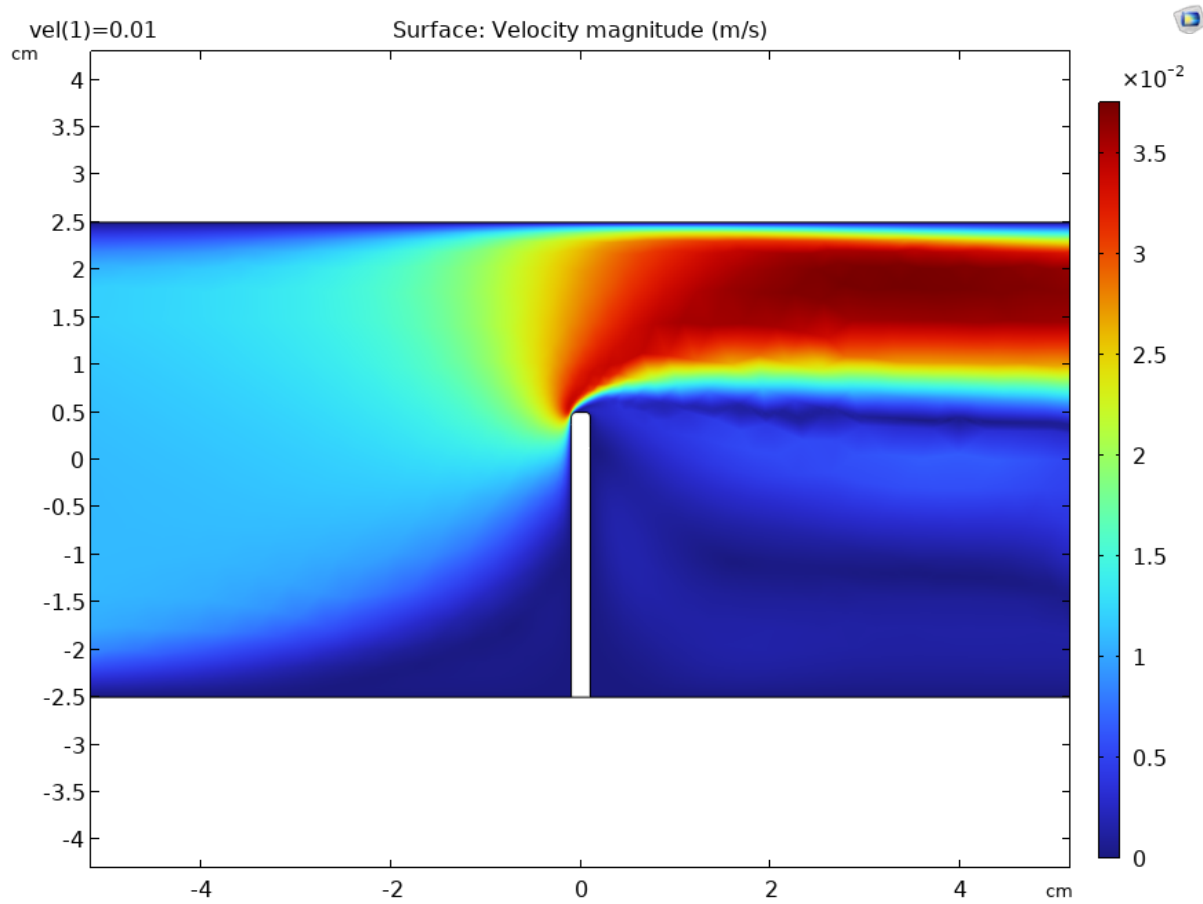


Figure 8: Velocity contour for $v=0.01$ and $E=1e8$

Velocity (m/s)	Displacement magnitude (cm)			
	Point 4	Point 5	Point 6	Point 8
0.01	0.000 074 248	0.000 075 844	0.000 075 843	0.000 074 246

Table 7: Displacement magnitudes at different points for velocity 0.01 m/s.

3.3.2 $v = 0.5 \text{ m/s}$

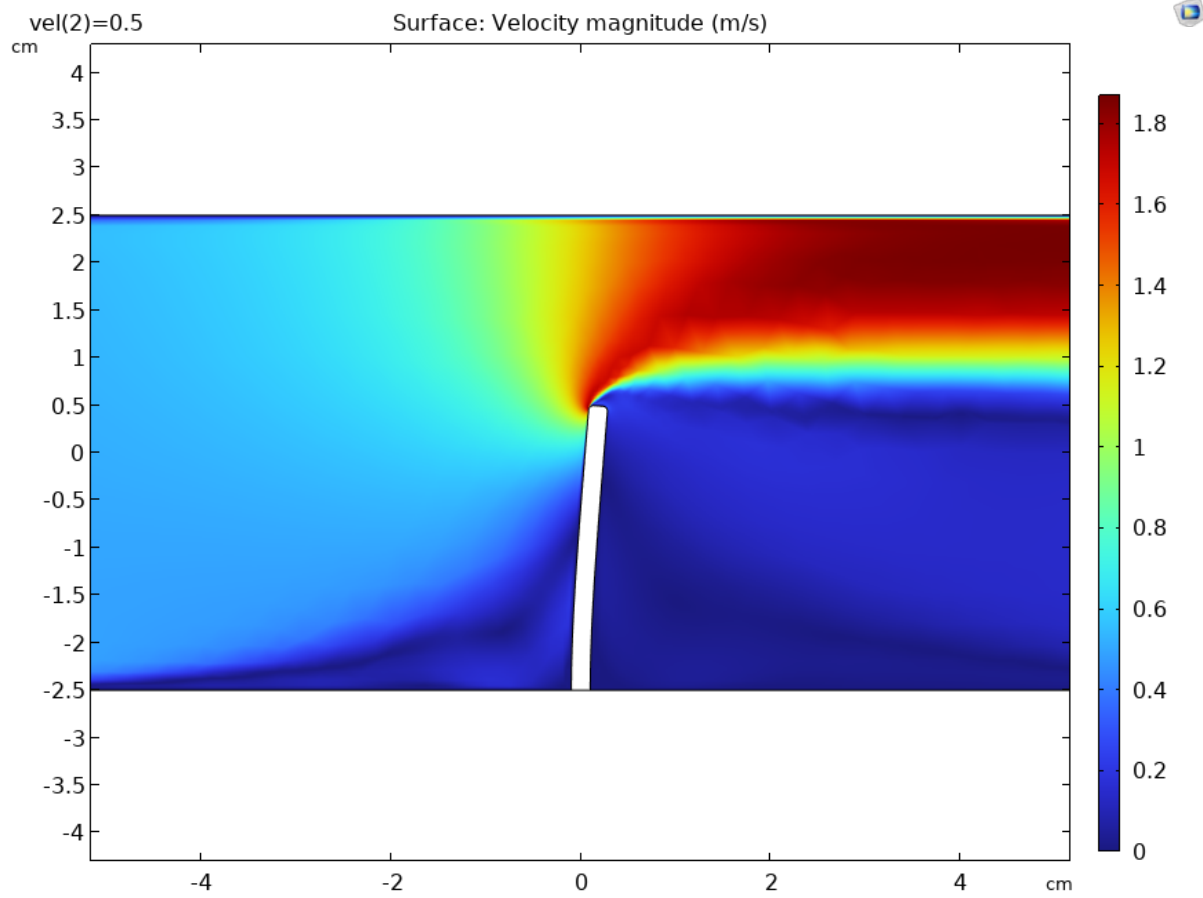


Figure 9: Velocity contour for $v=0.5$ and $E=1e8$

Velocity (m/s)	Displacement magnitude (cm)			
	Point 4	Point 5	Point 6	Point 8
0.5	0.182 29	0.186 18	0.186 14	0.182 20

Table 8: Displacement magnitudes at different points for velocity 0.5 m/s .

3.3.3 $v = 0.8 \text{ m/s}$

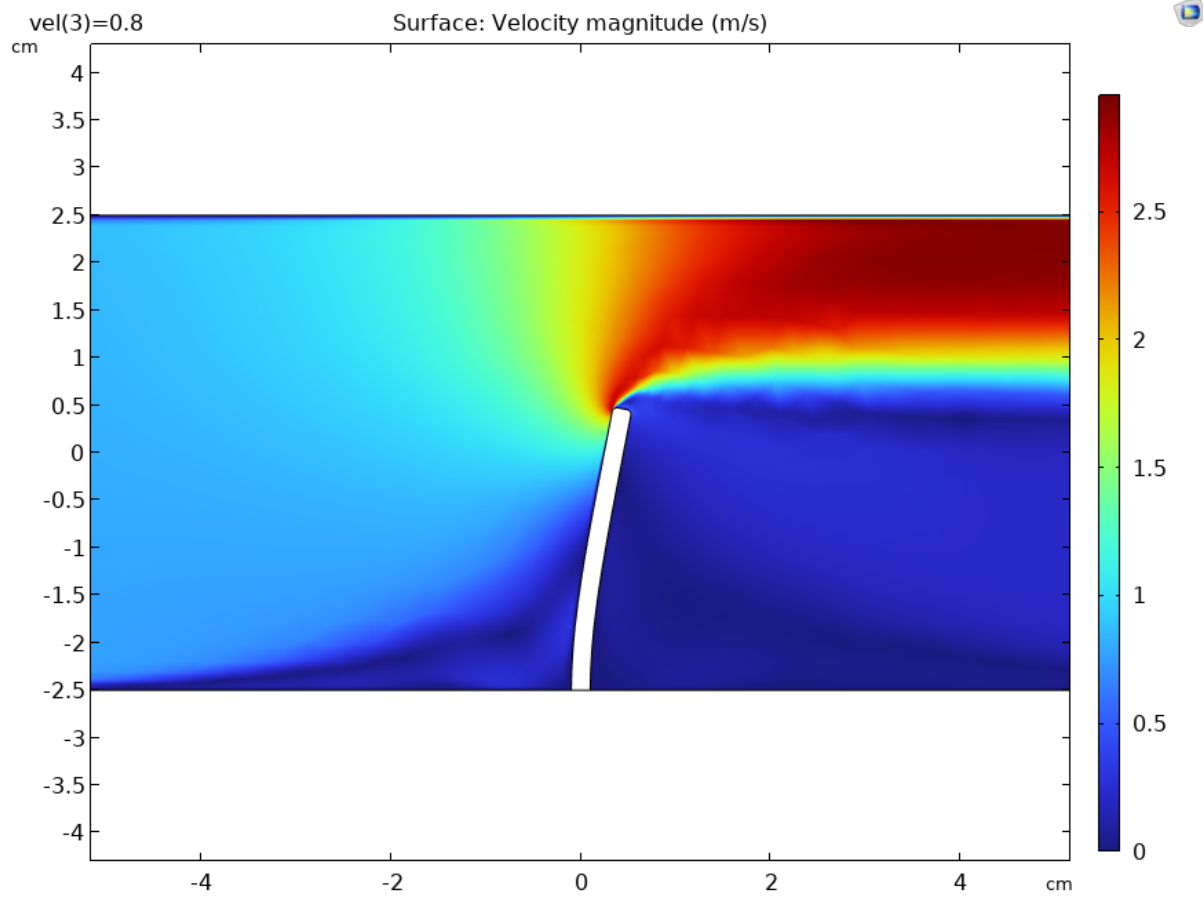


Figure 10: Velocity contour for $v=0.8$ and $E=1e8$

Velocity (m/s)	Displacement magnitude (cm)			
	Point 4	Point 5	Point 6	Point 8
0.8	0.431 64	0.440 77	0.440 53	0.431 16

Table 9: Displacement magnitudes at different points for velocity 0.8 m/s .

4 Mesh Details

For **Case 1** ($E = 1e6$), we use **normal** element size, but for **Case 2** and **Case 3**, we use **coarse** element size because with normal mesh size, COMSOL couldn't converge the answer despite using fully coupled or other method.

Also we use **Quartic Lagrange** method for Displacement field in all 3 cases because in this way we could converge the answers.